

Homework 8

Connor Mooneyhan

December 1, 2025

Disclaimer: AI was used for some question clarification and for some LaTeX help.

1. Let k be a field. Prove that the only two-sided ideals of the ring $R = M_n(k)$ of $n \times n$ matrices over k are R and $\{0\}$.

Proof. Note that $\{0\}$ and R are ideals of any ring R , so we need only prove that any nontrivial ideal is equal to R .

Define $E_{i,j}$ to be the matrix whose (i, j) entry is 1 where all other entries are 0. Note that right-multiplying a matrix by $E_{i,i}$ “chooses” the i^{th} row of the matrix in the sense that it yields a matrix whose i^{th} row is the same as the original matrix but with zeroes everywhere else. Similarly, left-multiplication by $E_{j,j}$ chooses the j^{th} column. We also adopt the notation $a \cdot 1_R$ for any $a \in k$ to mean the diagonal matrix with a in each nonzero entry.

Now let I be an ideal of R and let $A \in I$ be nonzero, choose i and j so that $A_{i,j}$ is nonzero. Then

$$E_{j,j}AE_{i,i}((A_{i,j})^{-1} \cdot 1_R) = E_{i,j},$$

so $E_{i,j} \in I$.

It is a basic fact from linear algebra that given any k, ℓ between 1 and n , we can conjugate by a permutation matrix P to achieve

$$PE_{i,j}P^{-1} = E_{k,\ell},$$

so $E_{k,\ell} \in I$. This allows us to see that since

$$1_R = \sum_{1 \leq k \leq n} E_{k,k},$$

1_R is in I , and thus $I = R$. □

2. Let S be a subset of a ring R . Define the left annihilator of S by

$$\text{leftAnn}_R(S) = \{r \in R \mid rs = 0 \text{ for all } s \in S\}.$$

(a) Show that $\text{leftAnn}_R(S)$ is a left ideal.

(b) Show that if S is a left ideal of R , then $\text{leftAnn}_R(S)$ is a two-sided ideal.

Proof.

- (a) Let $r \in \text{leftAnn}_R(S)$, and let $a \in R$. Then $(ar)s = a(rs) = a(0) = 0$, so $ar \in \text{leftAnn}_R(S)$ as desired.
- (b) Again, let $r \in \text{leftAnn}_R(S)$, and let $a \in R$. Then $(ra)s = r(as) = 0$ since S being a left ideal guarantees that $as \in S$. Thus $\text{leftAnn}_R(S)$ is also a right ideal, so it is a two-sided ideal.

□

3. Let G be a finite group, and R a ring. Suppose that $\mathcal{K} = \{k_1, \dots, k_n\}$ is a conjugacy class of G . Prove that $K = k_1 + k_2 + \dots + k_n$ is in the center of the group ring RG .

Proof. Given any $g \in G$, define $\varphi : G \rightarrow G$ by $a \mapsto gag^{-1}$. If $1 \leq i \leq n$, then $\varphi(k_i) = k_j$ for some j , and in fact $\varphi(\mathcal{K}) = \mathcal{K}$ since $\mathcal{K}$ is a conjugacy class. Thus

$$\begin{aligned} gKg^{-1} &= g \left(\sum_{i=1}^n k_i \right) g^{-1} \\ &= \sum_{i=1}^n gk_i g^{-1} \\ &= \sum_{j=1}^n k_j \\ &= K. \end{aligned}$$

Thus $gK = Kg$.

Now given any element $\sum_{i=1}^m r_i g_i$ of RG , we have

$$\begin{aligned} K \left(\sum_{i=1}^m r_i g_i \right) &= \sum_{i=1}^m r_i (Kg_i) \\ &= \sum_{i=1}^m r_i (g_i K) \\ &= \left(\sum_{i=1}^m r_i g_i \right) K. \end{aligned}$$

Thus K commutes with every element of RG , so it is in the center. □

4. Let R be a commutative ring, and let I and J be ideals of R . We say I and J are *comaximal* if $I + J = R$. Prove that if I and J are comaximal, then the rings $R/(I \cap J)$ and $R/I \times R/J$ are isomorphic via the (well-defined!) map given by $r + (I \cap J) \mapsto (r + I, r + J)$.

Proof. Let $\varphi : R/(I \cap J) \rightarrow R/I \times R/J$ be the map in question. We first need to show that it is a ring homomorphism. We have

$$\begin{aligned} \varphi((a + I \cap J) + (b + I \cap J)) &= \varphi((a + b) + I \cap J) \\ &= ((a + b) + I, (a + b) + J) \\ &= ((a + I) + (b + I), (a + J) + (b + J)) \\ &= (a + I, a + J) + (b + I, b + J) \\ &= \varphi(a + I \cap J) + \varphi(b + I \cap J), \end{aligned}$$

and similarly,

$$\begin{aligned}
\varphi((a + I \cap J)(b + I \cap J)) &= \varphi(ab + I \cap J) \\
&= (ab + I, ab + J) \\
&= ((a + I)(b + I), (a + J)(b + J)) \\
&= (a + I, a + J)(b + I, b + J) \\
&= \varphi(a + I \cap J)\varphi(b + I \cap J).
\end{aligned}$$

Also, $\varphi(1 + I \cap J) = (1 + I, 1 + J)$. Thus φ is a ring homomorphism.

Now assume that for some $a, b \in R$, $\varphi(a + I \cap J) = \varphi(b + I \cap J)$. Then

$$\begin{aligned}
(a + I, a + J) &= (b + I, b + J) \\
\implies (a - b + I, a - b + J) &= (I, J) \\
\implies a - b + I &= I \text{ and } a - b + J = J \\
\implies a - b \in I \text{ and } a - b \in J \\
\implies a - b \in I \cap J \\
\implies a - b + I \cap J &= I \cap J \\
\implies a + I \cap J &= b + I \cap J,
\end{aligned}$$

so φ is injective. If $c, d \in R$, then there exist $i_1, i_2 \in I$ and $j_1, j_2 \in J$ such that $c = i_1 + j_1$ and $d = i_2 + j_2$. Then

$$\begin{aligned}
(c + I, d + J) &= (i_1 + j_1 + I, i_2 + j_2 + J) \\
&= (j_1 + i_1 + I, i_2 + j_2 + J) \\
&= (j_1 + I, i_2 + J).
\end{aligned}$$

We now see that

$$\begin{aligned}
\varphi(i_2 + j_1 + I \cap J) &= (i_2 + j_1 + I, i_2 + j_1 + J) \\
&= (j_1 + i_2 + I, i_2 + j_1 + J) \\
&= (j_1 + I, i_2 + J) \\
&= (c + I, d + J).
\end{aligned}$$

Since c and d were arbitrary, we have surjectivity of φ , and thus φ is a ring isomorphism. $\square$

5. Let R be a commutative ring, and let S be a multiplicatively closed set of nonzero elements that contains 1. That is, if $s, s' \in S$, then $ss' \in S$. Let A be the set of ‘fractions’:

$$A = \left\{ \frac{r}{s} \mid r \in R \text{ and } s \in S \right\}.$$

Declare two fractions to be equivalent if

$$\frac{r}{s} \sim \frac{r'}{s'} \iff t(rs' - r's) = 0 \text{ for some } t \in S.$$

- (a) Prove that equivalence of fractions forms an equivalence relation. How can we simplify the equivalence relation if R is a domain?
- (b) The set of fractions with denominators in S modulo equivalence is denoted $S^{-1}R$. Show that $S^{-1}R$ is a commutative ring with identity under the operations

$$\frac{r}{s} + \frac{r'}{s'} = \frac{rs' + r's}{ss'} \quad \frac{r}{s} \frac{r'}{s'} = \frac{rr'}{ss'}.$$

Note that this includes showing that the multiplication and addition operations are well-defined.

- (c) Show that the function $\psi : R \rightarrow S^{-1}R$ defined by $\psi(r) = \frac{r}{1}$ is a ring homomorphism.
- (d) Compute the kernel of ψ (see Problem 1 on this homework).
- (e) Suppose that $\varphi : R \rightarrow T$ is a ring homomorphism such that $\varphi(s)$ is a unit for all $s \in S$. Show that there is a unique ring homomorphism $\varphi' : S^{-1}R \rightarrow T$ so that $\varphi = \varphi'\psi$.
- (f) State the previous result in terms of a universal property of the homomorphism $\psi : R \rightarrow S^{-1}R$ in the category of rings.

Proof.

- (a) Let $a, b, c \in R$ and $x, y, z \in S$. Then $ax - ax = 0$, so $a/x \sim a/x$. Also, if $t(ay - bx) = 0$, then $-t(bx - ay) = 0$, so $a/x \sim b/y \implies b/y \sim a/x$. Finally, if $t_1(ay - bx) = t_2(bz - cy) = 0$, then we just need to prove transitivity.

Note that the equivalence condition implies that $rs' - r's$ is a zero divisor, but if R is a domain, this means $rs' - r's = 0$, so we can simplify the condition to just $rs' = r's$.

(b)

- (c) If $a, b \in R$, then $\psi(1) = 1/1$, which is the identity in $S^{-1}R$,

$$\begin{aligned} \psi(a) + \psi(b) &= \frac{a}{1} + \frac{b}{1} \\ &= \frac{a \cdot 1 + b \cdot 1}{1 \cdot 1} \\ &= \frac{a + b}{1} \\ &= \psi(a + b), \end{aligned}$$

and

$$\begin{aligned} \psi(a)\psi(b) &= \frac{a}{1} \frac{b}{1} \\ &= \frac{ab}{1} \\ &= \psi(ab), \end{aligned}$$

so ψ is a ring homomorphism.

- (d) We are looking for $r \in R$ such that $\psi(r) = r/1 \sim 0/1$, i.e. $0 = t(r \cdot 1 - 0 \cdot 1) = tr = rt$ for some $t \in S$. It seems like the prompt is trying to get me to say that $\ker \psi = \text{leftAnn}_R(S)$ in the language of Problem 2, but elements of $\text{leftAnn}_R(S)$ need to annihilate *every* element of S , so this is not necessarily true. What I can say using that language is

$$\ker \psi = \bigcup_{s \in S} \text{leftAnn}_R(\{s\}).$$

(e)

(f)

□